Estimated Time of Arrival for Sustainable Transport Using Deep Neural Network

Aditya and Hina Firdaus

Abstract Time of arrival estimation is a challenge in a world where swiftness and accuracy are a new normal. During the year 1997, a summit in the United Nations showed concern for sustainable transport. This sustainability is for rural–urban linkage, pollution-free environment, health, etc. We understood the importance of estimating the time of arrival for the sustainable transportation and formulated a deep neural network model which increases our estimation of the time of heavy vehicles from source to destination. In our proposed research paper, we created a grid-based network dataset from a raw GPS truck data stored in a form of Data Lake. We introduced a vanilla neural network with three-layered architecture which works well with our grid-based dataset. Gradually, the velocity and volume of data increased and the dataset obtained was with labeled attributes of continuous type. The traditional machine learning algorithms failed or took more time to train on rapidly increasing GPS-dataset. The vanilla deep neural network in a three-layered architecture outperforms various deep learning method with an accuracy of 85%.

Keywords Neural network · Grid learning · Random forest · Logistic regression · Neural network · Sustainable transport

1 Introduction

Machine learning models are evolving rapidly, and so the data are increasing in velocity and volume. The industrialization is in revolution 4.0 where automation is mostly computerized. This situation [1] was well understood by United Nations in the year of 1997 and raised an issue for sustainable transport. The machine learning and big data can help us to understand the performance of exponentially increasing

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data size in various real-life applications such as health care, education, government, transport, etc.

1.1 Vanilla Neural Network

The concept of multilayer perceptron is based on forward and backward traversing [2]. The model is generally divided in three phases primary phase is input layer and tertiary layer is output. The output layer can have single to n number of probabilistic output. The middle layer is the hidden which encapsulates activation function and optimizers. This algorithm works well in the labeled dataset and falls between the [3] supervised learning and [4] unsupervised learning termed as [5] semi-supervised learning.

1.2 Ideation of Problem Statement

Literature of Albeaik et al. [6] have proposed deep neural network and deep reinforcement learning on a heavy truck archival dataset, where the performance of heavy truck is analyzed in various dynamics. The deep models performed better to understand the functionality of longitudinal truck structure. Nezafat et al. [7] also worked on the truck body dataset using transfer learning where the deep neural network hidden layer is treated to re-model the obtained image dataset of the truck. Here the emphasis was on pre-trained CNN models which improved the accuracy for classification of truck. Another paper we considered was by He et al. [8] here the bus journey travel time is predicted. The bus travel time includes traffic and halting time. The dataset they used was archival nature so long short term memory (LSTM) model was an able to predict accurately the arrival time of bus between sources to destination.

2 Methods and Materials

There were various challenges in the proposed project; firstly collection of heavy vehicle data precisely trucks. We contacted the local truck drivers with global positioning system (GPS). Raw data were collected in data lake, these data are stored in centralize location in the form of 'as is'. Data Lake is more advance and smooth transitioning between raw and processed data. We can store *N* numbers of structured, semi-structured, and unstructured data type of audio, videos, images, and texts. Despite of having lots of benefit using Data Lake, there are few challenges which we have faced, as the data size was increasing handling metadata got difficult. Second challenge we faced was ability to update and delete data, which make it less secure.

Table 1 Attributes of initial data collected

S. No.	Parameters	Total values
1	GPS_DATE	2,037,096
2	VEH_EXIT	
3	ACTUAL_ARRIVAL	
4	TIME_TAKEN	
5	IN_SEC	
6	PRD	
7	MONTH	
8	DAY	
9	HOUR	
10	DATEOFWEEK	
11	FROM	
12	TO	
13	DISTANCE	
14	TIME_DIFF	
15	HOUR	

The parameters in which our data were stored are in Table 1, where the attributes are GPS_DATE (date obtained from truck GPS), VEH_EXIT (vehicle exit time), ACTUAL_ARRIVAL (time of truck arrival and continuous type of data), TIME_TAKEN (encoded value in 0, 1, 2, 3, 4, 5, discrete type), IN_SEC (GPS data from truck in seconds discrete type), PRD (continuous type), MONTH (time series type), DAY (time series type), HOUR (time series type), DATEOFWEEK (time series type), FROM (continuous type), TO (continuous type), DISTANCE (continuous type), TIME_DIFF (continuous type), and HOUR (continuous type).

The major challenge in front of us is to predict the total duration of the trip done by the truck vehicle at the start of trip. We also need to re-calculate these estimated time of arrival predictions at every step of the process. This challenge is important from the technical point of view and also for the sustainability of the environment. If our solution is in a way where time of travel is less during the duration of journey which make the estimation of petrol/CNG/diesel price more accurate which help the truck owners to know the total estimation price of the journey in the beginning state itself.

Keeping these challenges in our mind, we made a huge deal focusing on arranging and pooling our raw data. These data are utilized in two ways, i.e., batch preprocessing and near real-time processing [9]. In one of the research paper, the collected GPS data from vehicle is only focused to the local roads, environment, and weather. These kinds of data do not work in general situation. The dataset we made will work for any individual or a company situated across the globe. That is why, we have not considered the route of roads from Google Maps or other source as the route changes a lot in due to development which can be external factors outside of our control.

Table 2 Attributes obtained after preprocessing

S. No.	Attributes	Total value
1	FROM_LAT	2,021,023
2	FROM_LON	
3	TO_LAT	
4	TO_LON	
5	MONTH	
6	DAYOFMONTH	
7	HOUR	
8	DAYOFWEEK	
9	DISTANCE	
10	ACTUAL_HOURS	
11	PREDICTED_HOURS	
12	DIS_CLUSTER	

As the data were from multiple streams, we divided the data processing into two different stages first was batch pre-processing. From Table 1, you can see the volume of our data was 2,037,096. With batching, we attained a refined dataset as shown in Table 2, FROM_LAT (latitude of source as continuous data type), FROM_LON (longitude of source as continuous data type), TO_LAT (latitude of destination as continuous data type), TO_LON (longitude of destination as continuous data type), MONTH (discrete type of data), DAYOFMONTH (discrete type of data), HOUR (discrete type of data), DAYOFWEEK (discrete type of data), DISTANCE (obtained attributes from source and destination distance as continuous type), ACTUAL_HOURS (obtained data of continuous type), PREDICTED_HOURS (obtained attributes of continuous type), and DIS_CLUSTER (obtained attributes of continuous type).

In the later stage, the preprocessed data are in near real-time processing, which help us analyzing the everyday data and evaluating them in almost real time.

The project was developed in Python programming language, ML library using TensorFlow library. For backend, we used structured data and we used oracle, and for unstructured, we used MongoDB. We have used eight core processor Linux servers.

For analyzing the data, we have introduced grid-based learning. Where the whole dataset is divided in grids and all the other grids are connected with each other. We were familiar with the real world problem of transport when heavy vehicles have to halt and take unusual turns as they may create an outliers in our analysis.

3 Model Development Evolution

In the previous section, we have seen the challenges of dataset creation and utilization. Now in this section, we will elaborate you about the process of model evaluation. The obtained dataset was first evaluated using the machine learning model, as you know the dataset was already obtained labeled. We choose supervised learning model like logistic regression to analyze the continuous type of data, but the algorithm could not improve the accuracy of more than 65% of total predicted time. We also found the model is not performing well if the batch size of data is increasing with higher volume, accuracy of the model started decreasing.

Aftermath, we choose random forest which showed an accuracy of 78% which was better than that of the logistic regression. Challenge we started facing when the size of data started increasing which made prediction to decline and training time was very high. Carefully, analyzing the machine learning model and their failure, we opted for neural network [10]. This model helped as training model with more than with 80% accuracy also with high volume of data. Refer Fig. 1, here we have modeled the whole process.

Primary stage of development is data collection and preprocessing which we have already explained in the previous section. Later stage is training of model in various machine learning and vanilla neural network. After that, we evaluated the model in the metrics of accuracy. The accuracy is maintained using MAPE metric model evaluation and for testing, we used A/B testing. Most researchers do not show the phase of deployment and monitoring of the model. Here we have deployed our trained model for automation in a regularly trained model set using the Rest API and hosted in Linux server. The use of live monitoring of the near real-time processing and trained data helped us to gain insight of the data. As in maintenance phase, we live compared the time of arrival with the previous accuracy metrics we obtained judging on various factors like halt, turns, etc.

In the process of creating the training model in vanilla neural network, we created a three-layered architecture. Each layer had its importance; first layer is input with the trained model second layer we had activation function with an optimizer [11]. Adam optimizer with [12] gradient descent optimizer. The third layer is output layer; we have a single output because the dataset we used is of continuous type. As shown

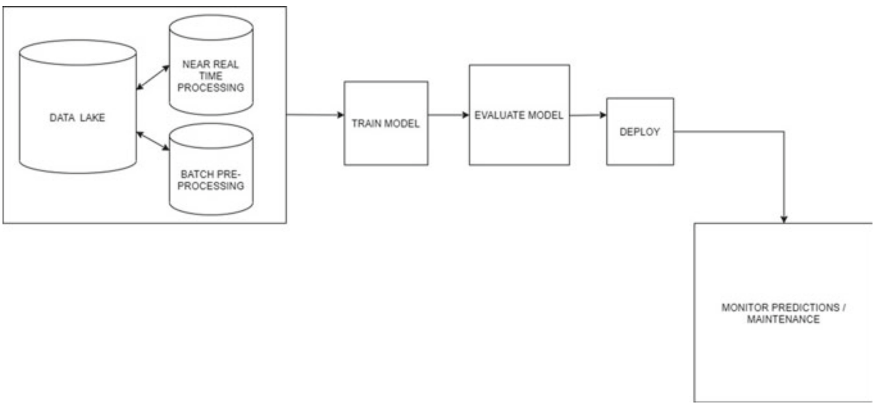

Fig. 1 Model for estimation of arrival time of truck

in the Fig. 2, the first input layer constitute of three major units that are, 1. truck live location (x_i), 2. distance between to and from location, (d_i). 3. time of arrival (a_i), and 4. truck type. The arrival of vehicle $x_i = [x_1, x_2, x_3, \ldots, x_n]$ each input calculates a speculated vehicle arrival by

$$x_i = a_i - d_i$$

where

a_i actual time of arrival
d_i departure time of the vehicle.

In the hidden layer, we have optimizers in a form of Adam optimizers and gradient descent optimizer with the activation function. Adam optimizer is referring

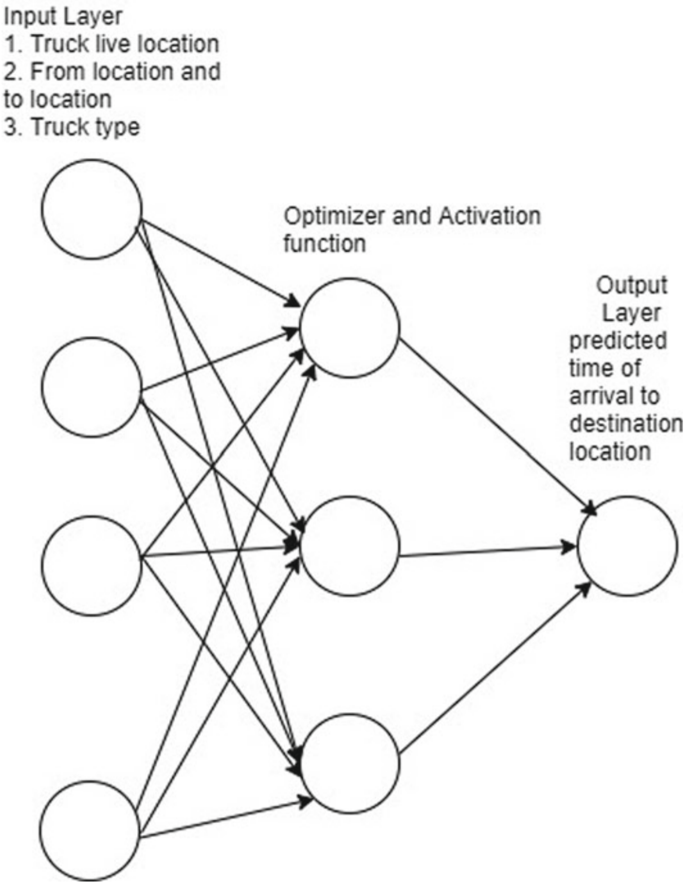

Fig. 2 Three-layered vanilla neural network architecture

as learning technique which updates rate for individual parameters. It uses successive and previous gradients and updates weight of each neural network. For the calculation of momentum, gradient descent algorithm is taken into consideration. This can be calculated as

$$W_{t+1} = W_t - LG_t$$
$$G_t = MG_{t-1} + (1 - M)[\delta f/\delta W_t]$$

where

G_t aggregate of gradients at time t
$G_{t\text{-}1}$ aggregate of gradients at time $t - 1$
W_t weights at time t
W_{t+1} weights at time $t + 1$
L learning rate at time t
M moving average parameter constant 0.9
δf derivative of loss function
δW_t derivative of weights at time t.

After the calculation of momentum, it is important to understand the formulation of adaptive gradient which we define as root mean square prop aka RMSprop.

$$W_{t+1} = W_t - L_t/(S_t + \varepsilon)^{1/2}[\delta f/\delta W_t]$$
$$S_t = MS_t + (1 - M)[\delta f/\delta W_t]^2$$

where

G_t aggregate of gradients at time t
W_t weights at time t
W_{t+1} weights at time $t + 1$
L learning rate at time t
M moving average parameter constant 0.9
δf derivative of loss function
δW_t derivative of weights at time t
ε a small positive constant (10^{-8})
S_t sum of square of past gradients.

The obtained Adam optimizers from the above calculation is

$$G_t = M_1 G_{t-1} + (1 - M_1)[\delta f/\delta W_t] S_t = M_2 S_t + (1 - M_2)[\delta f/\delta W_t]^2$$

M_1 and M_2 are decay rates of average gradients in the both methods where $M_1 = 0.9$ and $M_2 = 0.999$.

It is evident the Adam optimizer works fast in real-time fast moving or changing data because the rate of training is very low and performance measurement is very high.

The dataset we have used is of continuous in nature. The logistic regression traditionally should have performed well but due to the volume of data we are training regression is unable to perform. The third layer is output layer which act as regression model producing single output of the predicting the time of arrival for the truck at the location.

In the evaluation, we used simple A/B testing because we need to make a model which can enhance the prediction in every new input. This testing method works well in controlled random scenarios. In the A part of the test, we can keep the instances unchanged; on the other hand, the B part can be a new set of inputs. This is a hypothetical testing method so in various scenarios were considered and the best scenario is finally adapted for the model.

To curb any error in the model, we have used mean absolute percentage error (MAPE) which is used to calculate the exact forecast as an absolute average of error in percentage for each time period.

$$M = 1/N \sum_{t=1}^{N} [A_t - F_t/A_t]$$

where

N number of exact points
A_t actual elements
F_t forecast elements.

The algorithm we curate considering all these models is like this

Input:

Live locations of truck
Distance between from location and to location
Type of truck.

Output:

Predicted arrival time to the destination location

Step 1: Get the Current Location of the vehicle.
Step 2: Get the body type of a vehicle.
Step 3: Obtain the current Vehicle location Time(UTC/Date-time format).
Step 4: Split the datetime into 'Month', 'DayofMonth', 'Hour', 'Dayofweek'.
Step 5: Get the geo-location of To Branch.
Step 6: Label encoding body-type of vehicle.
Step 7: Calculate Distance between locations using Haversine formula.
Step 8: Train the data in vanilla neural network.

Table 3 Table for model accuracy

S. No.	Models	Accuracy (%)
1	Logistic regression	65
2	Random forest	78
3	Three-layered vanilla neural network	85

Step 9: Output will be the predicted hours as the estimated time of truck arrival.

4 Observation

The experiment we have performed was for a live project we did for the local truck vendors. The biggest challenge we have faced in the whole experimental phase was obtaining and storing the data. Another challenge was increasing the prediction of accuracy. Refer Table 3 for comparison between various models that we have used.

We can make an observation despite of using the best performing and state-of-the-art supervised learning model which generally outperforms in continuous type of data, failed in the scenario of our dataset [13] Which make it true the no lunch theorem in data science is more accurate which enable a thought to use more state-of-the-art algorithm on more complicated dataset to analyze the formulation and performance of the model. The deep learning is kind of semi-supervised learning which we have tried in our dataset. The addition of our own route and start time does not created issue in the accuracy of the model, which is a proof that the vanilla neural network being a classical gradient descent approach can work well in continuous huge volume data.

The monitoring and maintenance phase of our modulation was another challenge. Due to change in the route, the rest API use to get stuck and also the real time updating due to traffic was also not a factor that we have considered initially in our project. As we were using the GPS, the halting time in traffic or the other cases were purely neglected. This could be a challenge which we would like to take up in our rest research.

5 Concluding Remarks

In the proposed research work, we have used the entire classical machine learning model and re-evaluated their performance based on the dataset we have used. In the future, we are going to integrate the IoT and instead of near real-time processing, we will upgrade the proposed research for real-time locations. The biggest challenge is availability of hardware, because the biggest challenge for any machine learning model or neural network is type of dataset and volumes. We need more research and quality work where obtained dataset is mostly formulated in state-of-the-art algorithm to understand the behavior of an algorithm based on these factors. In the future work, we would like to use convolutional neural network, reinforcement neural network with our dataset.

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